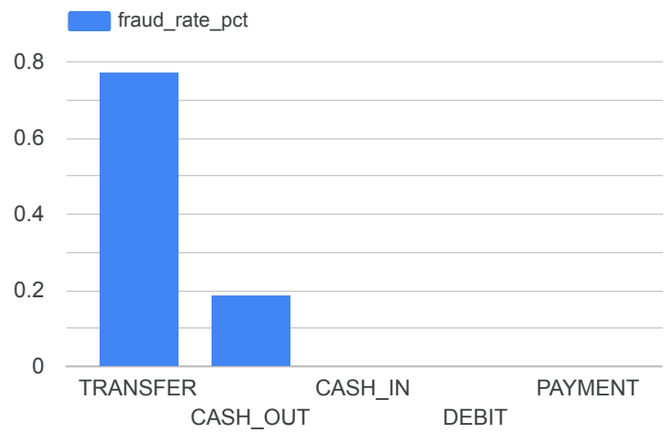
Fraud & Anomaly Detection — PaySim Transaction Analysis

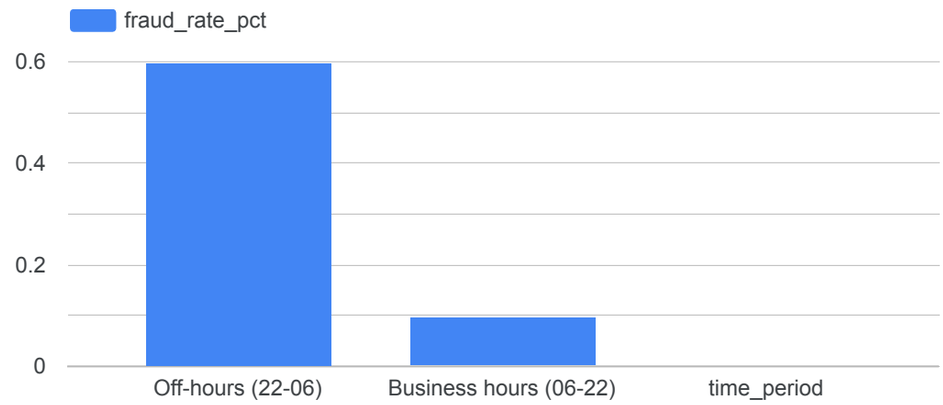
Author:Simran Anand | Data: Kaggle PaySim Synthetic Financial Transactions | Tool: SQL (SQLite), Looker Studio

Total Transactions	Total Confirmed Fraud	Total Flagged	Precision Rate
6,362,620	8,213	14,602	13.15

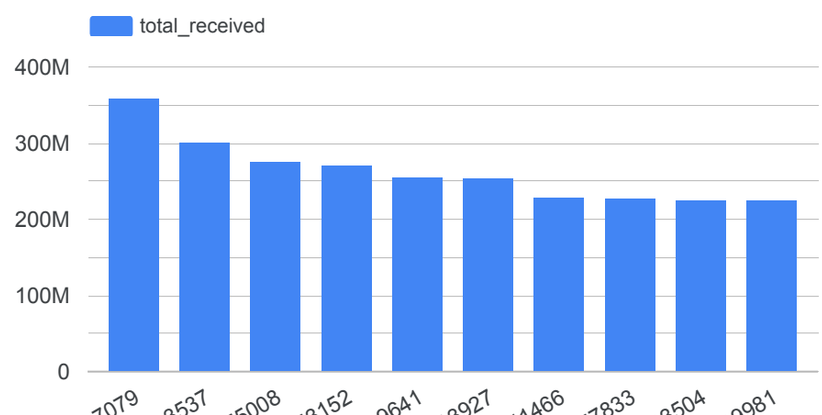
Fraud rate by transaction type



Off-hours transactions are 6.4x more likely to be fraudulent



Top 25 recipients by total transaction value received



Anomaly Breakdown — Timing and Payee Analysis

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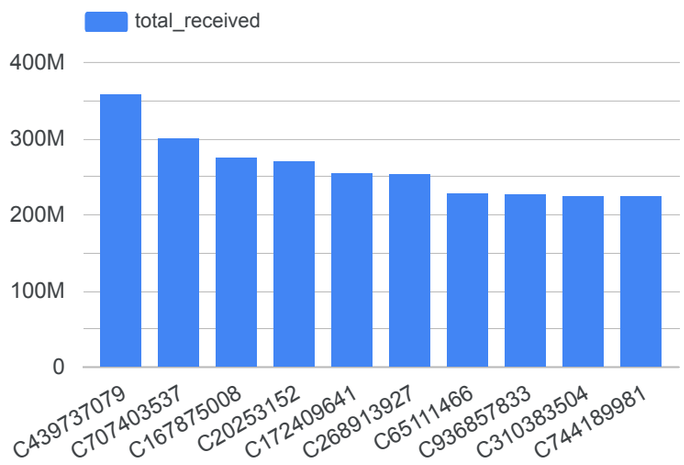
Off-hours fraud rate
0.6

Business hours fraud rate
0.09

Off-hours transactions are 6.4x more likely to be fraudulent despite representing only 7% of total volume

Top 25 recipients by total value received



Methodology & Recommended Controls

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Detection methodology and recommended controls per anomaly type

	Anomaly Type	Detection Method	Recommended Control	Flagged	Precision Rate
1.	Off-Hours Timing	Transactions between 22:00–06:00 (step % 24)	Require step-up authentication for TRANSFER and CASH_OUT transactions initiated between 22:00–06:00	448,417	0.01
2.	Large Transaction Outlier	Z-score > 3 standard deviations above type mean	Automated hold on TRANSFER and CASH_OUT transactions exceeding 3 standard deviations above mea…	14,602	0.13
3.	Payee Concentration	Top 25 recipients by total value received	Enhanced due diligence monitoring on any recipient receiving more than 1% of monthly transaction volume	25	0
4.	Near-Duplicate Payment	Same sender, recipient, amount ±1, within 24 hours	Implement duplicate payment detection rule in production transaction monitoring system	0	null

Limitations: This analysis was conducted on a synthetic dataset (PaySim). Thresholds were pre-defined prior to data exploration to avoid confirmation bias. All thresholds should be calibrated against historical confirmed fraud cases before production deployment.